

● EASY

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

04

10 answers · Rationale-rich · Teach from doc

Q1**D****D.** 329

Choice D is correct. It's given that the width of this rectangle is 7 centimeters and that the length of this rectangle is 40 centimeters longer than the width. Therefore, the length of this rectangle is $7 + 40$, or 47, centimeters. The area of a rectangle can be found by multiplying its length and its width. Therefore the area, in square centimeters, of this rectangle is $7 \cdot 47$, or 329.

Choice A is incorrect. This is the width, in centimeters, not the area, in square centimeters, of this rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2**D****D.** 24

Choice D is correct. The sum of the angle measures of a triangle is 180° . It's given that the measure of angle Q is 132° , the measure of angle P is 24° , and the measure of angle R is x° . It follows that $x + 132 + 24 = 180$, or $x + 152 = 180$. Subtracting 152 from both sides of this equation yields $x = 24$.

Alternate approach: It's given that in the triangle shown, $PQ = QR$. Thus, triangle PQR is an isosceles triangle. Since base angles of an isosceles triangle are congruent, the measure of angle P is equal to the measure of angle R . Therefore, the measure of angle R is 24° , and the value of x is 24.

Choice A is incorrect. This is the sum of the measures, in degrees, of angles Q and P , not the measure, in degrees, of angle R .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures, in degrees, of angles P and R , not the measure, in degrees, of angle R .

Q3**B**

B. $\frac{35}{171}$

Choice B is correct. The sine of an acute angle in a right triangle is the ratio of the length of the side opposite that angle to the length of the hypotenuse. The hypotenuse of a right triangle is the side opposite the right angle. In right triangle ABC , side BC is the side opposite angle A and side AB is the hypotenuse. It's given that the length of side BC is 35 units and the length of side AB is 171 units. Therefore, the value of $\sin A$ is $\frac{35}{171}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the ratio of the length of the hypotenuse to the length of the side opposite angle A rather than the ratio of the length of the side opposite angle A to the length of the hypotenuse.

Choice D is incorrect. This is the length of the hypotenuse rather than $\sin A$.

Q4**D**

D. 1,568

Choice D is correct. The area, A , of a rectangle can be found using the formula $A = lw$, where l represents the length and w represents the width of the rectangle. It's given that a rectangle has a length of 56 inches and a width of 28 inches. Substituting 56 for l and 28 for w in the formula $A = lw$ yields $A = (56)(28)$, or $A = 1,568$. Therefore, the area, in square inches, of the rectangle is 1,568.

Choice A is incorrect. This is the width, in inches, not the area, in square inches, of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q5**D****D.** 147

Choice D is correct. It's given that line m is parallel to line n , and line t intersects both lines. It follows that line t is a transversal. When two lines are parallel and intersected by a transversal, exterior angles on the same side of the transversal are supplementary. Thus, $x + 33 = 180$. Subtracting 33 from both sides of this equation yields $x = 147$. Therefore, the value of x is 147.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6**A****A.** 7

Choice A is correct. The area A , in square meters, of a rectangle is the product of its length l , in meters, and its width w , in meters; thus, $A = lw$. It's given that a rectangle has an area of 63 square meters and a length of 9 meters. Substituting 63 for A and 9 for l in the equation $A = lw$ yields $63 = 9w$. Dividing both sides of this equation by 9 yields $7 = w$. Therefore, the width, in meters, of the rectangle is 7.

Choice B is incorrect. This is the difference between the area, in square meters, and the length, in meters, of the rectangle, not the width, in meters, of the rectangle.

Choice C is incorrect. This is the square of the length, in meters, not the width, in meters, of the rectangle.

Choice D is incorrect. This is the product of the area, in square meters, and the length, in meters, of the rectangle, not the width, in meters, of the rectangle.

Q7**B****B.** 45°

Choice B is correct. It's given that triangles EFG and JKL are congruent such that angle E corresponds to angle J . Corresponding angles of congruent triangles are congruent, so angle E and angle J must be congruent. Therefore, if the measure of angle E is 45° , then the measure of angle J is also 45° .

Choice A is incorrect. This is the measure of angle K , not angle J .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**880**

The correct answer is 880. The volume, V , of a right rectangular prism is given by the formula $V = lwh$, where l is the length, w is the width, and h is the height of the prism. It's given that a right rectangular prism has a length of 11 meters, a width of 8 meters, and a height of 10 meters. Substituting 11 for l , 8 for w , and 10 for h in the formula $V = lwh$ yields $V = (11)(8)(10)$, or $V = 880$. Therefore, the volume, in cubic meters, of the prism is 880.

Q9**47**

The correct answer is 47. Based on the figure, the angle with measure x° and the angle with measure 133° together form a straight line. Therefore, these two angles are supplementary, so the sum of their measures is 180° . It follows that $x + 133 = 180$. Subtracting 133 from both sides of this equation yields $x = 47$.

Q10

986

The correct answer is 986. The area, A , of a rectangle is given by $A = lw$, where l is the length of the rectangle and w is its width. It's given that the length of the rectangle is 34 centimeters (cm) and the width is 29 cm. Substituting 34 for l and 29 for w in the equation $A = lw$ yields $A = 3429$, or $A = 986$. Therefore, the area, in square centimeters, of this rectangle is 986.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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